CLAIMS

1. Wheat grain (*Triticum aestivum*) comprising an embryo, an endosperm, starch and and a reduced level or activity of total starch branching enzyme II (SBEII) protein, wherein the embryo comprises a loss of function mutation in each of 5 to 12 alleles of endogenous *SBEII* genes selected from the group consisting of *SBEIIa*-A, *SBEIIa*-B, *SBEIIa*-D, *SBEIIb*-A, *SBEIIb*-B and *SBEIIb*-D, such that the level or activity of total SBEII protein in the grain is between 2% and 30% of the level or activity of total SBEII protein in a wild-type wheat grain, said 5 to 12 alleles including 4, 5 or 6 *SBEIIa* alleles each comprising a loss of function mutation, wherein when the number of *SBEIIa* alleles comprising a loss of function mutation is only 4 then the number of *SBEIIb* alleles comprising a loss of function mutation is 6, and wherein the grain comprises an amylose content of at least 50% (w/w) as a proportion of the total starch of the grain.

2. The wheat grain of claim 1, wherein

a) the embryo of the grain is homozygous for alleles in each of 2 or 3 *SBEIIa* genes, each of the homozygous alleles comprising a loss of function mutation,

b) the grain comprises an allele which comprises a partial loss of function mutation which expresses an SBEIIa or SBEIIb enzyme which in amount and/or activity corresponds to between 2% and 60%, or between 10% and 50%, of the amount or activity of the corresponding wild-type allele,

c) the number of null alleles of *SBEIIa* genes in the embryo is 2 or 4,

d) the 4, 5 or 6 *SBEIIa* alleles each comprising a loss of function mutation are each null alleles,

e) the embryo has no null alleles of *SBEIIb* genes, or the number of null alleles of *SBEIIb* genes in the embryo is 2, 4 or 6,

f) the embryo comprises two *SBEIIb* alleles each comprising a partial loss of function mutation,

g) the grain comprises null alleles of the *SBEIIa* gene on the A genome, B genome, D genome, A and B genomes, A and D genomes, or B and D genomes,

h) the grain comprises null alleles of the *SBEIIb* gene on the A genome, B genome, D genome, A and B genomes, A and D genomes, B and D genomes, or all three of the A, B and D genomes,

i) the grain comprises only one or only two SBEIIb proteins which have starch branching enzyme activity when produced in developing endosperm, or only one or only two SBEIIb proteins which are detectable by Western blot analysis,

j) the grain comprises a null mutation which is a deletion mutation in the A, B or D genome which deletes at least part of an *SBEIIa* gene and at least a part of an *SBEIIb* gene on that genome,

k) the grain comprises a null mutation which is a deletion mutation in the A, B or D genome which deletes the whole of the *SBEIIa* gene and/or the whole of the *SBEIIb* gene on that genome,

l) the grain comprises a null mutation in an *SBEIIa* gene which is an amino acid substitution mutation,

m) the embryo comprises a null mutation in an *SBEIIa* gene, or null mutations in more than one *SBEIIa* gene, wherein each mutation is independently selected from the group consisting of a deletion mutation, an insertion mutation, a splice-site mutation, a premature translation termination mutation and a frameshift mutation,

n) the grain has only one SBEIIa protein as determined by Western blot analysis, wherein said protein is encoded by one of the *SBEIIa-A*, *SBEIIa-B* and *SBEIIa-D* genes and has reduced starch branching enzyme activity when produced in developing wheat endosperm when compared to an SBEIIa protein encoded by the corresponding wild-type gene, or

o) a combination of more than one of a) to n).

3. The grain of claim 1 or claim 2, in which the level or activity of total SBEII protein in the grain is between 2% and 15%, or between 3% and 10%, or between 2% and 20% or between 2% and 25%, of the level or activity of total SBEII protein in the wild-type wheat grain.

4. The grain of any one of claims 1 to 3, in which the amount or activity of SBEIIa protein in the grain is less than 2%, or between 2% and 15%, or between 3% and 10%, or between 2% and 20% or between 2% and 25%, of the amount or activity of SBEIIa protein in the wild-type wheat grain.

5. The grain of any one of claims 1 to 4, comprising an amylose content of at least 60% (w/w) or at least 67% (w/w) as a proportion of the total starch of the grain.

6. The grain of any one of claims 1 to 5, which is non-transgenic or is free of any exogenous nucleic acid that encodes an RNA which reduces expression of an *SBEIIa* gene.

7. The grain of any one of claims 1 to 6, which has a germination rate of between about 70% and about 100% relative to the germination rate of a wild-type grain.

8. The grain of any one of claims 1 to 7, wherein the level or activity of total SBEII protein is determined by assaying the SBEII protein in grain while it is developing in a wheat plant or by assaying the level or activity of SBEII protein by immunological means.

9. The grain of any one of claims 1 to 8, wherein the starch of the grain is characterised by one or more properties selected from the group consisting of:
   (i) comprising at least 2% resistant starch,
   (ii) comprising a reduced glycaemic index (GI),
   (iii) comprising reduced level of amylopectin,
   (iv) comprising distorted starch granules,

(v) reduced starch granule birefringence,
(vi) reduced swelling volume,
(vii) modified chain length distribution and/or branching frequency,
(viii) delayed end of gelatinisation temperature and increased peak temperature,
(ix) reduced viscosity,
(x) increased molecular weight of amylopectin,
(xi) a modified percentage of starch crystallinity, and
(xii) a modified percentage crystallinity of A-type or B-type starch,

relative to wild-type wheat starch granules or starch.

10. The grain of any one of claims 1 to 9, which is processed so that it is no longer capable of germinating, or which is kibbled, cracked, par-boiled, rolled, pearled, milled or ground grain.

11. The grain of any one of claims 1 to 10, which is capable of producing a wheat plant which is male and female fertile.

12. A wheat plant (*Triticum aestivum*) which produces grain, the grain comprising an embryo, an endosperm, starch and a reduced level or activity of total SBEII protein, wherein the embryo comprises a loss of function mutation in each of 5 to 12 alleles of endogenous *SBEII* genes selected from the group consisting of *SBEIIa*-A, *SBEIIa*-B, *SBEIIa*-D, *SBEIIb*-A, *SBEIIb*-B and *SBEIIb*-D, such that the level or activity of total SBEII protein in the grain is between 2% and 30% of the level or activity of total SBEII protein in a wild-type wheat grain, said 5 to 12 alleles including 4, 5 or 6 *SBEIIa* alleles each comprising a loss of function mutation, wherein when the number of *SBEIIa* alleles comprising a loss of function mutation is only 4 then the number of *SBEIIb* alleles comprising a loss of function mutation is 6, wherein the grain comprises an amylose content of at least 50% (w/w) as a proportion of the total starch of the grain, and wherein the wheat plant is male and female fertile.

13. Wheat wholemeal or flour produced from the grain of any one of claims 1 to 11, the wholemeal or flour comprising a reduced level or activity of total SBEII protein and a loss of function mutation in each of 5 to 12 alleles of endogenous *SBEII* genes selected from the group consisting of *SBEIIa*-A, *SBEIIa*-B, *SBEIIa*-D, *SBEIIb*-A, *SBEIIb*-B and *SBEIIb*-D, such that the level or activity of total SBEII protein in the wholemeal or flour is between 2% and 30% of the level or activity of total SBEII protein in a wild-type wheat wholemeal of flour, said 5 to 12 alleles including 4, 5 or 6 *SBEIIa* alleles each comprising a loss of function mutation, wherein when the number of *SBEIIa* alleles comprising a loss of function mutation is only 4 then the number of *SBEIIb* alleles comprising a loss of function mutation is 6, and wherein the starch of the wholemeal or flour has an amylose content of at least 50% (w/w).

14. The wholemeal or flour of claim 13 wherein the starch of the wholemeal or flour is characterised by one or more of:
    (i) comprising at least 60% (w/w), or at least 67% (w/w) amylose,
    (ii) comprising at least 2% resistant starch,
    (iii) comprising a reduced glycaemic index (GI),
    (iv) comprising reduced level of amylopectin,
    (v) comprising distorted starch granules,
    (vi) reduced starch granule birefringence,
    (vii) reduced swelling volume,
    (viii) modified chain length distribution and/or branching frequency,
    (ix) delayed end of gelatinisation temperature and higher peak temperature,
    (x) reduced viscosity,
    (xi) increased molecular weight of amylopectin,
    (xii) modified % crystallinity
    (xiii) modified percentage crystallinity of A-type or B-type starch,

relative to a wild-type wheat starch granules or starch.

15. A food product comprising a food ingredient at a level of at least 10% on a dry weight basis, wherein the ingredient is wheat grain of any one of claims 1 to 11 or the wholemeal or flour of claim 13 or claim 14.

16. A food product according to claim 15 wherein the resistant starch content is between 1% to 20%

17~~16~~. A method of producing starch, comprising the steps of i) obtaining wheat grain according to any one of claims 1 to 11, and ii) extracting the starch from the grain, thereby producing the starch.

18~~17~~. A method of screening a wheat plant or wheat grain, the method comprising (i) determining the amount or activity of SBEIIa and/or SBEIIb in a wheat plant or wheat grain relative to the amount or activity in a wild-type plant or grain and (ii) selecting a plant or grain having between 2% and 30% of the amount or activity of total SBEII protein in a wild-type plant or grain, wherein the selected plant or grain comprises a loss of function mutation in each of 5 to 12 alleles of endogenous *SBEII* genes selected from the group consisting of *SBEIIa*-A, *SBEIIa*-B, *SBEIIa*-D, *SBEIIb*-A, *SBEIIb*-B and *SBEIIb*-D, such that the level or activity of total SBEII protein in the grain is between 2% and 30% of the level or activity of total SBEII protein in the wild-type wheat grain, said 5 to 12 alleles including 4, 5 or 6 *SBEIIa* alleles each comprising a loss of function mutation, wherein when the number of *SBEIIa* alleles comprising a loss of function mutation is only 4 then the number of *SBEIIb* alleles comprising a loss of function mutation is 6, and wherein the grain comprises an amylose content of at least 50% (w/w) as a proportion of the total starch of the grain.

19~~18~~. A method of producing a food or a drink comprising the following steps: (i) obtaining grain according to any one of claims 1 to 11, (ii) processing the grain to produce a food or drink ingredient, and (iii) adding a food or drink ingredient from (ii) to another food or drink ingredient, thereby producing the food or drink.

~~19. A food product for use in improving one or more parameters of metabolic health, bowel health or cardiovascular health, or of preventing or reducing the severity or incidence of metabolic disease, bowel disease or cardiovascular disease in a subject, the food product comprising a food ingredient at a level of at least 10% on a dry weight basis, wherein the food ingredient is wheat grain of any one of claims 1 to 11, the wholemeal or flour of claim 13 or 14.~~

20. A method of producing grain, comprising the steps of i) obtaining a wheat plant according to claim 12, ii) harvesting wheat grain from the plant, and iii) optionally, processing the grain.